

$\mathbb{Z}^n = \mathbb{Z}$

$$U_\eta = T_\eta, \quad U_\xi = T_\xi$$

$$\begin{pmatrix} T_{\xi} & T_{\eta} \\ S_{\xi} & S_{\eta} \end{pmatrix} = \begin{pmatrix} \zeta_T & \zeta_S \\ \eta_T & \eta_S \end{pmatrix}^{-1}$$

$$= \begin{pmatrix} g_z & a \\ 0 & 1 \end{pmatrix}^{-1}$$

$$= \frac{1}{g_x} \begin{pmatrix} 1 & -a \\ 0 & g_x \end{pmatrix}$$

$$\therefore T_1 = \frac{-a}{g_1}$$

$$\tau_{\xi} = \frac{1}{g_{\xi}}$$

$$\begin{aligned} U_T &= T_T \\ &= -\frac{-a}{g_c} \\ &= -a T_s \\ &= -a U_s \end{aligned}$$

$u = \tau$ とすれば (u, v) は求める座標系

$$(\xi, \eta) : E(\xi, 0) = 1.$$

$$F(\xi, \eta) = 0.$$

$$G(\xi, 0) = G_y(\xi, 0) = 0.$$

$$G_{\eta\eta}(\xi, 0) \neq 0.$$

$$\Rightarrow \begin{cases} u = u(\xi, \eta) \\ v = h(\xi) \eta \end{cases} \quad (h(\xi) = B(\xi)^{\frac{1}{4}})$$

$$\begin{cases} a(\xi, \eta) u_\xi + u_\eta = 0 \\ u(\xi, 0) = \underbrace{\xi}_{f(\xi)} \end{cases}$$

$$B(\xi) := \frac{\tilde{G}\eta(\xi, 0)}{2}$$

$$F=0 \iff \frac{u_\xi v_\xi}{\tilde{E}} + \frac{u_\eta v_\eta}{\tilde{G}} = 0 \quad \left(\begin{array}{l} v_\xi = h'(\xi) \\ v_\eta = h(\xi) \end{array} \right)$$

$$\Leftrightarrow \frac{h'(\xi)}{\tilde{E}} u_\xi + \frac{h(\xi)}{\tilde{G}} u_\eta = 0. \Leftrightarrow \begin{cases} u_\eta = - \frac{\tilde{G}}{\tilde{E} h(\xi)} u_\xi \\ u(\xi, 0) = \xi. \end{cases}$$

$$= a(x, y)$$

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$$\begin{cases} \frac{d\xi}{d\tau} = a(\xi, \eta), \xi(0) = s \\ \frac{d\eta}{d\tau} = 1, \eta(0) = 0 \end{cases} \Rightarrow \eta(\tau) = \tau$$

ξ として $S = \tau TL$.

$$\begin{cases} \frac{d\xi}{d\tau} = a(\xi, \tau) \\ \xi(0) = s \end{cases} \Rightarrow \xi(\tau) = \xi^s(\tau) \quad \text{or} \quad \xi(s, \tau) \text{ とおく}$$

$\eta(\tau) = \tau$

$\xi(s, 0) = s$

$$(s, \tau) \xrightarrow{\varphi} (\xi(s, \tau), \eta(\tau))$$

local diffeo.

$$\xi_s(s, 0) = \frac{\partial}{\partial s}(\xi^s(0)) = \frac{\partial}{\partial s}(s) = 1 \neq 0, \quad \xi_\tau = \frac{\partial}{\partial \tau} \xi^s(\tau) = \frac{d\xi}{d\tau} = a$$

$\eta_s = 0, \eta_\tau = 1$

$\exists \varphi^{-1}$

$$(\xi, \tau) = \varphi^{-1}(\xi, \eta)$$

$$\begin{matrix} \eta=0 \\ \Downarrow \\ \tau=0 \end{matrix} \quad \text{--- ①}$$

Prop $U(\xi, \tau) := S(\xi, \eta)$ とおく

$$U_\eta = -a U_\xi \quad \text{--- ①}$$

$$U(\xi, 0) = \xi \quad \text{--- ②}$$

(proof) $\frac{\partial U}{\partial \tau}(\xi, 0) = S(\xi, 0) = \xi$

① ② ③

$$\begin{pmatrix} S_\xi & S_\eta \\ T_\xi & T_\eta \end{pmatrix} = \begin{pmatrix} \xi_s & \xi_\tau \\ \eta_s & \eta_\tau \end{pmatrix}^{-1} = \begin{pmatrix} \xi_s & a \\ 0 & 1 \end{pmatrix}^{-1} = \frac{1}{\xi_s} \begin{pmatrix} 1 & -a \\ 0 & \xi_s \end{pmatrix}$$

$$S_\xi = \frac{1}{\xi_s}, S_\eta = -\frac{a}{\xi_s}$$

$$\therefore U_\eta = S_\eta = -a S_\xi = -a U_\xi //$$

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$$\begin{cases} \frac{d\xi}{d\tau} = a(\xi, \eta), \xi(0) = s \\ \frac{d\eta}{d\tau} = 1, \eta(0) = 0 \end{cases} \Rightarrow \eta(\tau) = \tau$$

5.2.4E S は可逆.

$$\begin{cases} \frac{d\xi}{d\tau} = a(\xi, \tau) \\ \xi(0) = s \end{cases} \Rightarrow \xi(\tau) = \xi(s, \tau) \quad \text{!!} \quad \xi(s, \tau) \text{ とおく}$$

$\eta(\tau) = \tau \Rightarrow \xi(s, 0) = s$

$$(s, \tau) \xrightarrow{\varphi} (\xi(s, \tau), \eta(\tau))$$

local diffeo.

$$\begin{pmatrix} \xi_s(s, 0) = \frac{\partial}{\partial s}(\xi(s, 0)) = \frac{\partial}{\partial s}(s) = 1 \neq 0, \xi_\tau = \frac{\partial}{\partial \tau} \xi(s, \tau) = \frac{d\xi}{d\tau} = a \\ \eta_s = 0, \eta_\tau = 1 \end{pmatrix}$$

$\exists \varphi^{-1}$

$$(\xi(s, \tau), \eta(\tau)) = \varphi^{-1}(\xi, \eta)$$

$$\begin{matrix} \eta=0 \\ \updownarrow \\ \tau=0 \end{matrix} \quad \text{--- } \textcircled{0}$$

Prop $U(\xi, \eta) := S(\xi, \eta)$ とおく

$$\begin{cases} U_\eta = -a U_\xi \quad \text{--- } \textcircled{1} \\ U(\xi, 0) = \xi \quad \text{--- } \textcircled{2} \end{cases}$$

$$\textcircled{0} \text{ と } \xi(s, 0) = s \text{ より}$$

(proof) $\textcircled{2}$ より $U(\xi, 0) = S(\xi, 0) = \xi$

$$\begin{aligned} \textcircled{1} \text{ より } \begin{pmatrix} S_\xi & S_\eta \\ \tau_\xi & \tau_\eta \end{pmatrix} &= \begin{pmatrix} \xi_s & \xi_\tau \\ \eta_s & \eta_\tau \end{pmatrix}^{-1} \\ &= \begin{pmatrix} \xi_s & a \\ 0 & 1 \end{pmatrix}^{-1} = \frac{1}{\xi_s} \begin{pmatrix} 1 & -a \\ 0 & \xi_s \end{pmatrix} \end{aligned}$$

$$S_\xi = \frac{1}{\xi_s}, S_\eta = -\frac{a}{\xi_s}$$

$$\therefore U_\eta = S_\eta = -a S_\xi = -a U_\xi //$$

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$$\begin{cases} \frac{d\xi}{d\tau} = a(\xi, \eta), \xi(0) = s \\ \frac{d\eta}{d\tau} = 1, \eta(0) = 0 \end{cases} \quad \tau=0$$

$$\Rightarrow \eta(\tau) = \tau$$

5.2.14 E U は変換

$$\frac{d\xi}{d\tau} = a(\xi, \tau) \Rightarrow \exists! \xi(\tau) = \xi^u(\tau)$$

$$\xi(0) = u$$

$$\eta(\tau) = \tau \rightarrow \xi(u, \tau) = u$$

$$(u, \tau) \xrightarrow{\varphi} (\xi(u, \tau), \eta(\tau))$$

local diffeo.

$$\xi_u(u, 0) = \frac{\partial}{\partial u} (\xi^u(0)) = \frac{\partial}{\partial u} (u) = 1 \neq 0, \xi_\tau = \frac{\partial}{\partial \tau} \xi^u(\tau) = \frac{d\xi}{d\tau} = a$$

$$\eta_u = 0, \eta_\tau = 1$$

$\exists \varphi^{-1}$

$$(u, \tau) = \varphi^{-1}(\xi, \eta)$$

Prop 上で定めた $U(\xi, \eta)$ は

$$\begin{cases} U_\eta = -a U_\xi - 1 \\ U(\xi, 0) = \xi \end{cases}$$

(proof) ①と $\xi(u, 0) = u$ ②から従う

①証明

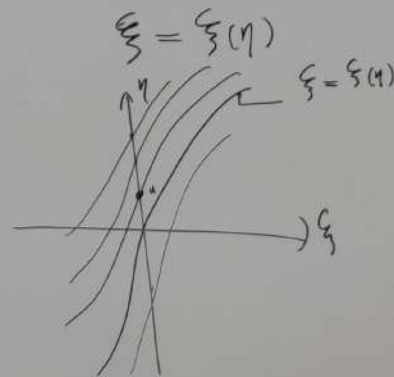
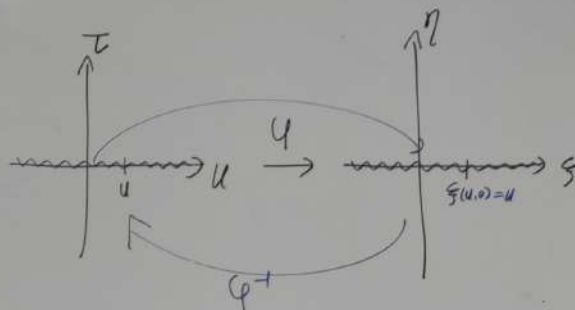
$$\begin{pmatrix} U_\xi & U_\eta \\ \tau_\xi & \tau_\eta \end{pmatrix} = \begin{pmatrix} \xi_u & \xi_\tau \\ \eta_u & \eta_\tau \end{pmatrix}^{-1}$$

$$= \begin{pmatrix} \xi_u & a \\ 0 & 1 \end{pmatrix}^{-1} = \frac{1}{\xi_u} \begin{pmatrix} 1 & -a \\ 0 & \xi_u \end{pmatrix}$$

$$U_\xi = \frac{1}{\xi_u}, U_\eta = -\frac{a}{\xi_u}$$

$$\therefore U_\eta = -a U_\xi - 1 //$$

$$\begin{matrix} \eta=0 \\ \Downarrow \\ \tau=0 \end{matrix} \quad \text{①}$$

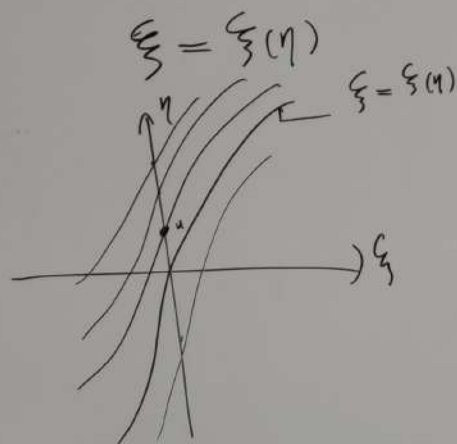
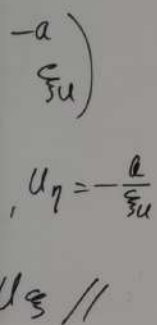


$(\xi, \eta) : E$

$$\Rightarrow \begin{cases} u = u(\xi, \eta) \\ v = h(\xi) \end{cases}$$

$$F=0 \Leftrightarrow$$

$$\Leftrightarrow$$



$$\begin{cases} Q(\xi, \eta) u_\xi + v_\eta = 0 \\ u(9, 0) = \xi, \\ \quad \quad \quad \Pi \\ \quad \quad \quad f(\xi) \end{cases}$$

$$\Rightarrow \begin{cases} u = u(\xi, \eta) \\ v = h(\xi) \eta \end{cases} \quad (h(\xi) := B(\xi)^{\frac{1}{4}})$$

$$(B(\xi) := \frac{\tilde{G}_{11}(\xi, 0)}{2})$$

$$F=0 \iff \frac{u_\xi \mathcal{D}_\xi}{\tilde{E}} + \frac{u_\eta \gamma_\eta}{\tilde{G}} = 0 \quad \begin{cases} V_\xi = h'(\xi) \\ V_\eta = h(\xi) \end{cases}$$

$$\begin{aligned} \Leftrightarrow \frac{u_\xi \mathcal{D}_\xi}{\tilde{E}} + \frac{u_\eta \gamma_\eta}{\tilde{G}} &= 0 & \begin{cases} V_\xi = h'(\xi) \\ V_\eta = h(\xi) \end{cases} & \quad \begin{matrix} \boxed{\frac{h'(\xi)}{\tilde{E}} \tilde{G}} \\ u_\xi \end{matrix} = a(\xi, \eta) \\ \Leftrightarrow \frac{h'(\xi)}{\tilde{E}} u_\xi + \frac{h(\xi)}{\tilde{G}} u_\eta &= 0 & \Leftrightarrow \begin{cases} u_\eta = - \frac{\boxed{\frac{h'(\xi)}{\tilde{E}} \tilde{G}}}{h(\xi)} u_\xi \\ u(\xi, 0) = \xi \end{cases} \end{aligned}$$

5/14 子真 関1 や子.

① p : non-deg. 1st kind

$$\Rightarrow \exists (u,v) \text{ s.t. } \begin{cases} E(u,0)=1, F(u,v)=0, \\ G(u,0)=G_v(u,0)=0, G_w(u,0)=2. \end{cases}$$

for $\vec{x}$

$$h(\vec{x}) = \left(\frac{G_w(\vec{x},0)}{2} \right)^{\frac{1}{2}}$$

② p : 退化双核 2. 1st kind

$$\Rightarrow \exists (u,v) \text{ s.t. } \begin{cases} E(u,0)=1, F(u,v)=0 \\ G = G_v = G_w = G_{vw} = 0, \text{ on } v=0. \\ G_{vw}(u,0)=b. \end{cases}$$

HW

① p : non-deg. 1st kind

$$\Rightarrow \exists (u, v) \text{ s.t. } \begin{cases} E(u, 0) = 1, F(u, v) = 0, \\ G(u, 0) = G_v(u, 0) = 0, G_w(u, 0) = 2. \end{cases}$$

封直

$$h(\frac{1}{2}) = \left(\frac{\tilde{G}_{uv}(1, 0)}{2} \right)^{\frac{1}{q}}$$

② p : 退化反例 2. 1st kind

$$\Rightarrow \exists (u, v) \text{ s.t. } \begin{cases} E(u, 0) = 1, F(u, v) = 0 \\ G = G_v = G_w = G_{vw} = 0, \text{ on } v=0. \\ G_{vw}(u, 0) = b. \end{cases}$$

HW